

ATM2/3 EVK Introduction

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This page features design information for Atmosic Evaluation Kits. Check the ATMx2xx Evaluation Kit Hardware Design Information to determine your hardware revision and contact your Atmosic representative if the information is not available from this page.

ATM2201 EVK (rev E2a)

Features: BLE, Wakeup RX, *QFN5x5*

- Schematic (Release Date August 10. 2022)
- Assembly Drawing (Release Date August 10, 2022)
- Full Design Download (Release Date August 10, 2022) [1 MB]

ATM2202 EVK (rev E2b)

Features: BLE, Wakeup RX, *QFN5x5*

- User Guide (Release Date April 14, 2021)
- Schematic (Release Date August 10, 2022)
- Assembly Drawing (Release Date August 10, 2022)
- Power consumption Evaluation (Release Date April 14, 2021)
- Full Design Download (Release Date August 10, 2022) [1 MB]

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Start From...

- EVK Guide:
  - 6x6: ATM3221,ATM2221
  - 5x5: ATM3201,ATM2201, ATM3202, ATM2202,
  - CSP:ATM2251, 7x7: ATM2231U
  - 7x7: ATM3330e
  - 5x5: ATM3325
- EVK User Guide: jumper configuration
- Interface Board\I/O adaptor board: V3, SWD Download, Debug Print, HCI/DTM Test
- SDK Quick Start Guide: Install RDI Interface, make REF_BCN:=power_profile run_all
- EVK Power Consumption Evaluation: Tx, Rx, Retention and Hibernation
- EVK Energy Harvesting Quick Start Guide: HARV1 SMA, HARV_OUT, Vstore

Further Materials: Customer Portal (**HW Design Guide, Reference Manual**)

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Power & Interface

USB1/UART1, is designated as the programming and UART1 debug interface

USB0/UART0, this port is used for HCI message communication



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Confirm the RDI Driver Ready

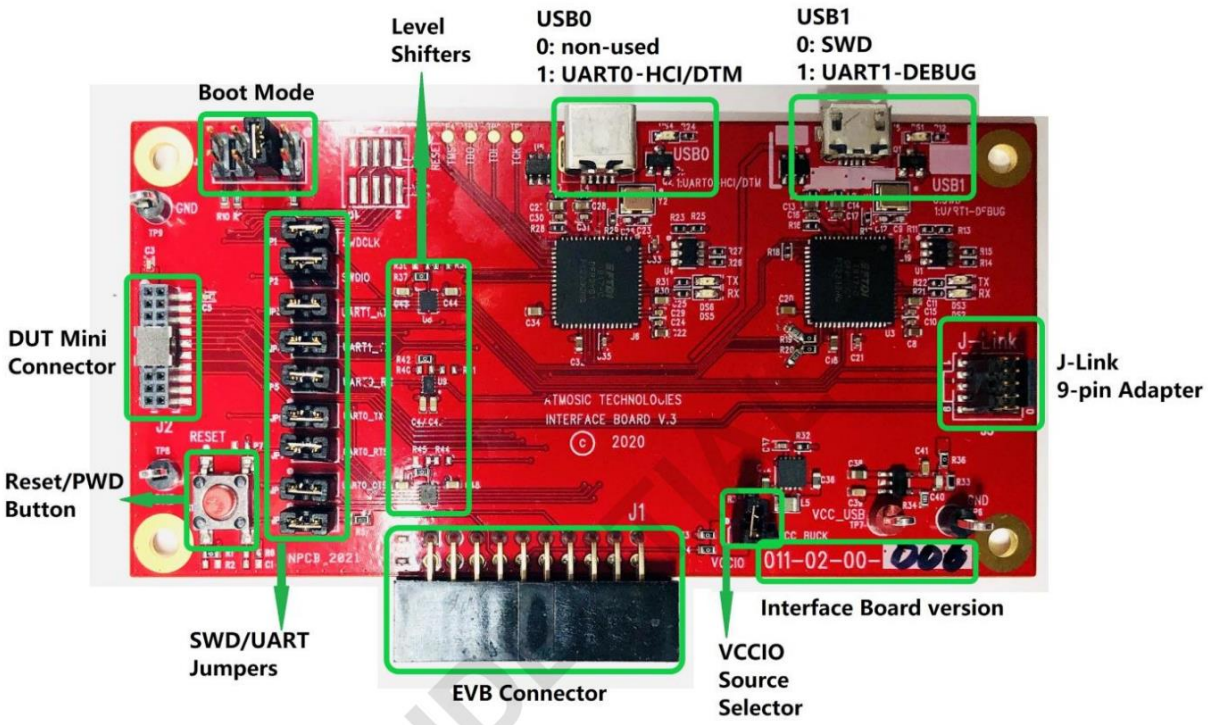
Any new Atmosic EVK (ATM2/3 & ATM33) will need reinstallation of Atmosic RDI USB driver.

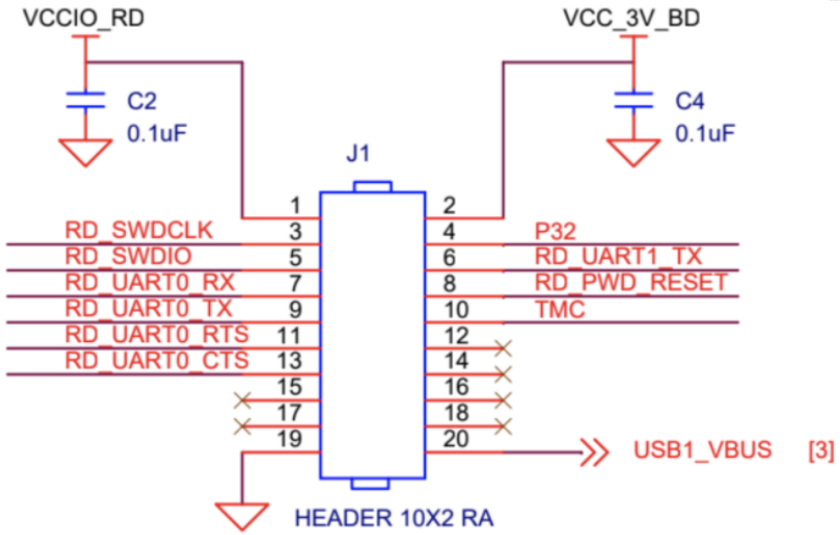


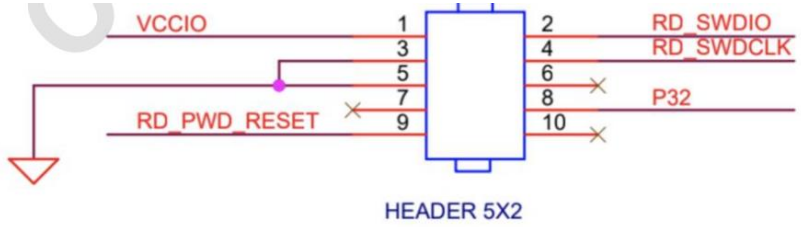
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Interface Board

- USB1: UART1_Tx, SWD
- USB0: UART0, 4-wire HCI or 2-wire DTM
- Mini connector: 1.27mm
- EVB connector: 2.54mm
- J-Link: 9-pin adapter



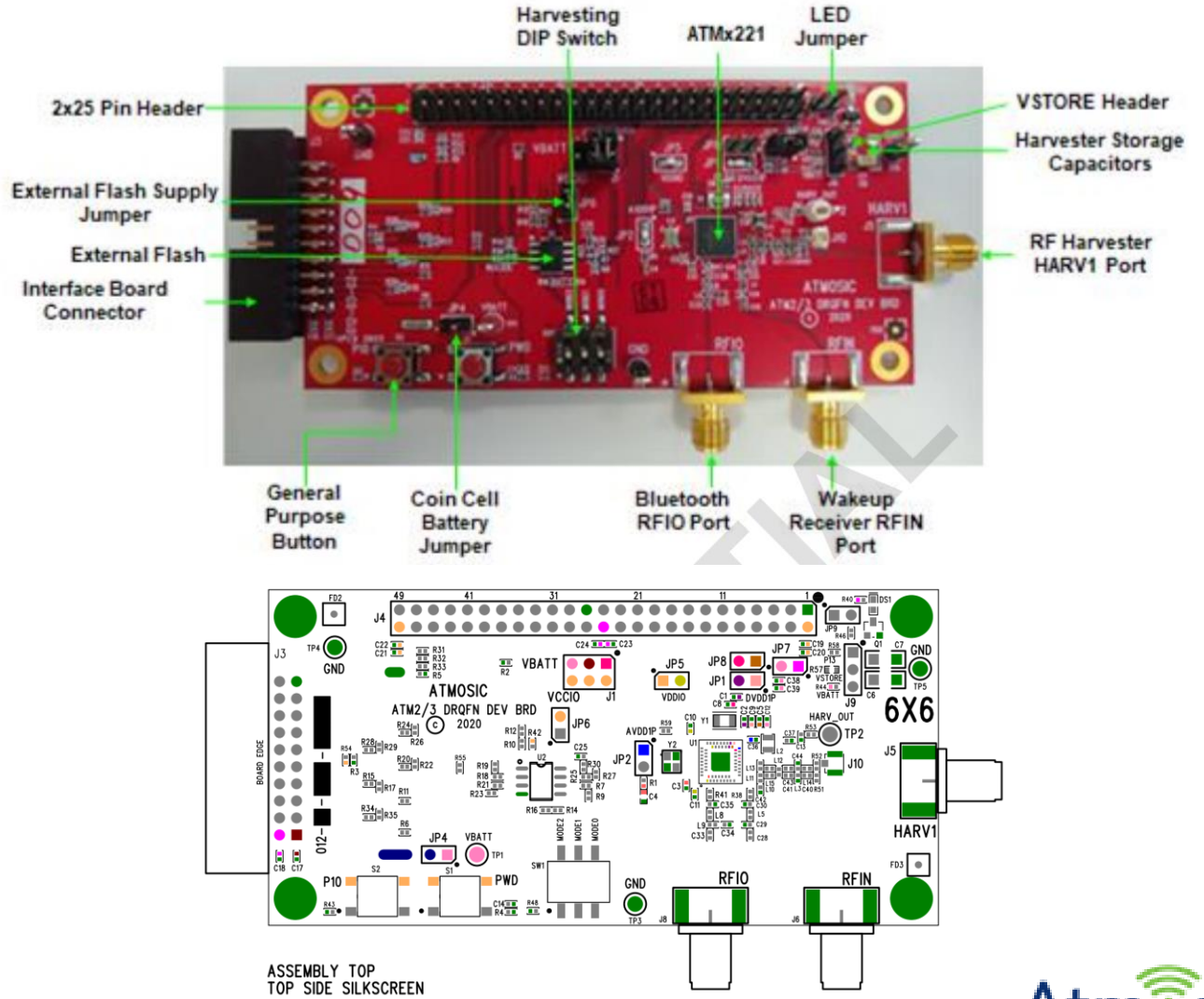




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ATM 3221 EVK Board

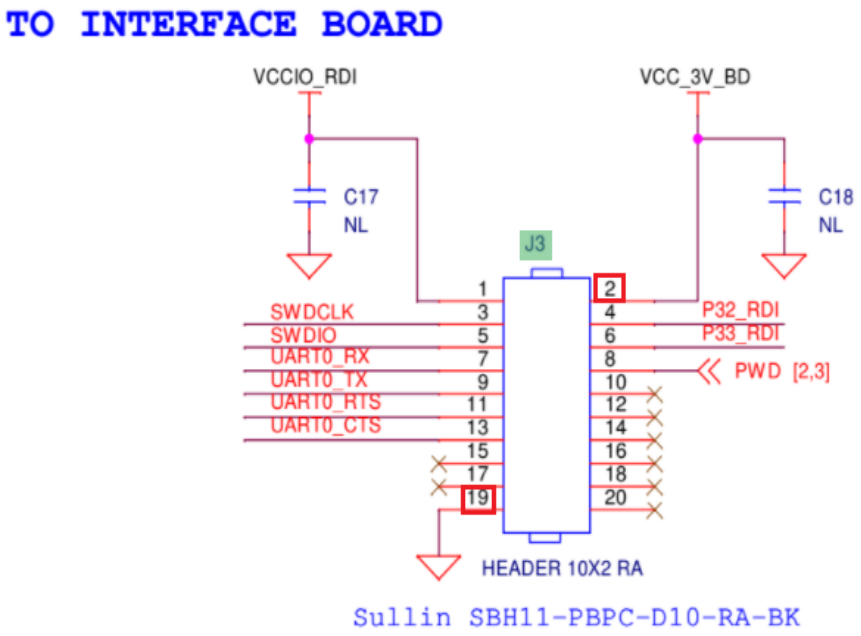
- Coin cell battery jumper
- EVB 20 pins connector
- 2x25 pin header
- External flash supply jumper
- Interface board connector
- Harvesting DIP switch
- Bluetooth RFIO port
- Wakeup receiver RFIN port
- RF harvester HARV1 port
- Harvester storage capacitors

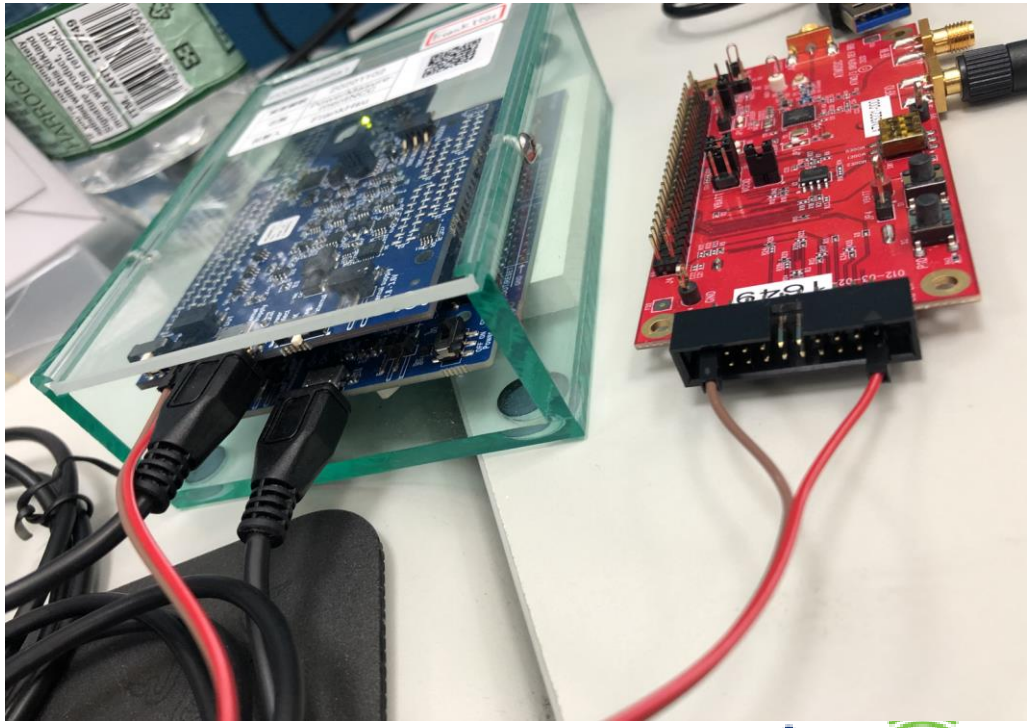


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ATM33 EVK Power Measurement

- AtmosicSDK\5.2.0\platform\atm3\ATM32xx-x1x\examples\BLE_adv\make REF_BCN:=power_profile run_all
- Connect the USB cable to the PPK II
- Connect 3V and GND to EVB





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RF PHY Testing Setup

- Attach an RF cable to RFIO and the Bluetooth tester
- HCI mode (4-wire): connect USB0 (UART0) to the PC, run the RFTool or a test scrip from the Bluetooth tester
- DTM mode (2-wire): connect USB0 (UART0) to the Bluetooth tester

Measured RF Parameter	Measured Value			Unit
	Min	Typical	Max	
Tx Power	2.5	3.5	4.5	dBm
Rx Sensitivity @ 1M	-92	-94	-96	dBm
Rx Sensitivity @ 2M	-90	-92	-94	dBm



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ADV. Packet Edit

Path : C:\AtmosicSDK\5.4.0.1\platform\atm2\ATM22xx-x1x\examples\BLE_adv
File name : makefile

```
include reference_beacons.mk
flash_nvds.data := $(reference_beacon_$(REF_BCN)) $(if $(AUTO_RESTART),09-APP_BLE_RSTRT_DUR/500ms)
```

reference_beacons.mk

REF_BCN := simple_beacon

```
# simple_beacon
#
# Sends scannable beacon forever without hibernation.
# This will allow mobile applications on iOS and Android to always see
# the device when scanning.
#
reference_beacon_simple_beacon := \
        $(supply_bd_addr) \
        06-APP_BLE_ACT_CRT_CMD/legacy_nc_s_100ms \
        0b-APP_BLE_ADV_DATA/simple_beacon \
        0c-APP_BLE_SCAN_RESP_DATA/simple_beacon \
        11-SLEEP_ENABLE/ret_drop \
        12-EXT_WAKEUP_ENABLE/enable \
```


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ADV. Packet Edit

0c-APP_BLE_SCAN_RESP_DATA/simple_beacon

simple_beacon.tds

simple_beacon.tds

파일 편집 보기

```
#0F                     # Length of Complete Local Name
#09                     # Complete Local Name
#41 74 6D 6F 73 69 63 20 42 65 61 63 6F 6E # 'Atmosic Beacon'

0B                      # Length of Complete Local Name
09                      # Complete Local Name
4C 53 54 20 42 65 61 63 6F 6E # 'LST Beacon'
```

0b-APP_BLE_ADV_DATA/simple_beacon

```
# Legacy ADV - no more than 28 bytes of payload

03      # Length of Service List
03      # Service List
aa fe   # Eddystone ID

0e                      # Length of Service Data
16                      # Service Data
aa fe                   # Eddystone ID
10                      # Frame type: URL
00                      # Power
01                      # https://www.
61 74 6d 6f 73 69 63    # "atmosic"
07                      # .com

#08                     # Length of Shortened Local Name
#08                     # Shortened Local Name
#41 74 6D 6F 73 69 63   # 'Atmosic'

04                      # Length of Shortened Local Name
08                      # Shortened Local Name
4C 53 54                # 'LST'
```


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ADV. Packet Edit

```
/c/AtmosicSDK/5.4.0.1/platform/atm2/ATM22xx-x1x/examples/BLE_adv
make: *** [/c/AtmosicSDK/5.4.0.1/platform/atm2/ATM22xx-x1x/user/nvds.mk:464: push_flash_nvds] Error 1

lian@DESKTOP-SQ681TD MINGW32 /c/AtmosicSDK/5.4.0.1/platform/atm2/ATM22xx-x1x/examples/BLE_adv
$ make run_all BOARD=ATMEVK_M2202
/c/AtmosicSDK/5.4.0.1/platform/atm2/ATM22xx-x1x/tools/bin/Windows_NT/nvds_tool  -b -t 06 -d tag_data/06-APP_BLE_AC
d tag_data/0b-APP_BLE_ADV_DATA/simple_beacon.bin -t 0c -d tag_data/0c-APP_BLE_SCAN_RESP_DATA/simple_beacon.bin -t
dog.bin -t 18 -d tag_data/18-PROG_DELAY/default.bin -t 2e -d tag_data/2e-SLEEP_ALGO_DUR/sleep_algo.bin > flash_nvd
FTDI_BENIGN_BOOT=1 FTDI_HARD_RESET=1 /c/AtmosicSDK/5.4.0.1/tools/openocd/bin/Windows_NT/openocd -s /c/AtmosicSDK/5
DK/5.4.0.1/platform/atm2/ATM22xx-x1x/openocd -f atm2x_openocd.cfg -c "init; release_reset; sleep 100; set_normal_b
Open On-Chip Debugger 0.11.0-00017-g131241a97 (2021-11-01-11:08)
Licensed under GNU GPL v2
For bug reports, read
        http://openocd.org/doc/doxygen/bugs.html
Using FTDI
Using Interface Board
Info : FTDI SWD mode enabled
get_coredump
Info : clock speed 1000 kHz
Warn : gdb services need one or more targets defined
/c/AtmosicSDK/5.4.0.1/tools/openocd/bin/Windows_NT/openocd -s /c/AtmosicSDK/5.4.0.1/tools/openocd/tcl -s /c/Atmosi
openocd -f atm2x_openocd.cfg -c "init; verify_rom_version; sydney_load_nvds flash_nvds.bin ; exit"
Open On-Chip Debugger 0.11.0-00017-g131241a97 (2021-11-01-11:08)
Licensed under GNU GPL v2
For bug reports, read
        http://openocd.org/doc/doxygen/bugs.html
Using FTDI
Using Interface Board
Info : FTDI SWD mode enabled
get_coredump
Info : clock speed 1000 kHz
Info : SWD DPIDR 0x0bb11477
Info : Sydney.cpu: hardware has 4 breakpoints, 2 watchpoints
Info : gdb port disabled
target halted due to debug-request, current mode: Thread
xPSR: 0xc1000000 pc: 0x00005530 msp: 0x20014000
target halted due to debug-request, current mode: Thread
xPSR: 0xc1000000 pc: 0x00005530 msp: 0x20014000
ATM2202-x1x silicon: 5x5 pkg, Stacked flash
GigaDevice 8Mb flash (1048576 bytes)
112 bytes written at address 0x10078000
downloaded 112 bytes in 0.023845s (4.587 KiB/s)
save layout info file in partition_info.map
```


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문서 활용

1. **Embedded Application Development AN**

3 System Utilities

3.1 Application State Machine

3.2 Log Utility

3.2.1 ATM_LOG Macro

3.2.2 Debug Log Level

3.2.3 How to Use

3.3 Power Management

3.3.1 Power Saving Modes

3.3.2 Power Mode Lock

3.3.3 Power Manager API

3.3.4 atm_pm lock example

3.4 Software Timer

3.4.1 Software Timer API

3.4.2 sw_timer example

3.5 Software Event

3.5.1 Software Event API

3.5.2 sw_event example

4. Hardware Drivers

4.1 GPIO Driver

4.1.1 atm_gpio Module

4.2 I2C Driver

4.2.1 i2c Module

4.3 SPI Driver

4.3.1 Pin Assignment for SPI

4.3.2 SPI Module

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